

Experiment 02: Raspberry Pi Initialization

Roll Number:21BIT191

Objective:

To initialize and configure a Raspberry Pi 4B by flashing Custom OS setting up VNC Viewer for desktop control

Equipment and Components:

Raspberry Pi 4B, External Drive (Micro SD/ Pen Drive), Power Adapter

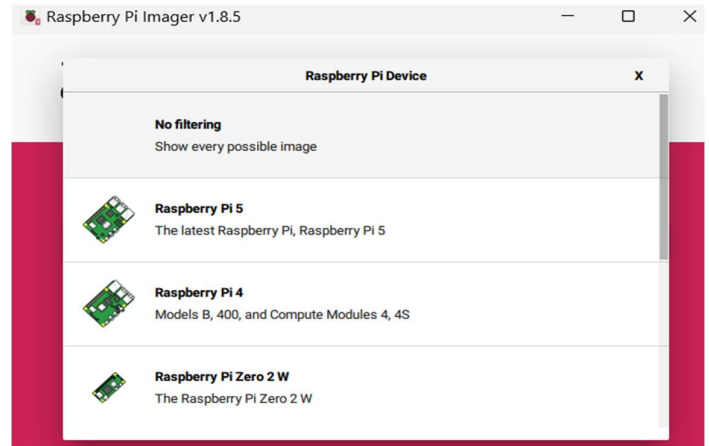
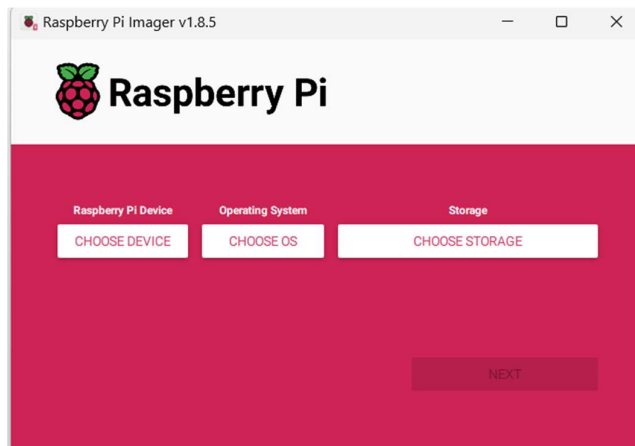
Theory:

The Raspberry Pi 4B is a versatile, small-sized. To set it up, you need an SD card or USB drive to hold the operating system (OS) and files, which are typically transferred using an SD card reader connected to a laptop. The Raspberry Pi Imager software is used to flash the OS onto the SD card or USB drive. This involves selecting the storage device, configuring settings like enabling SSH and setting up Wi-Fi, and then formatting and writing the OS image, which takes about 10-15 minutes.

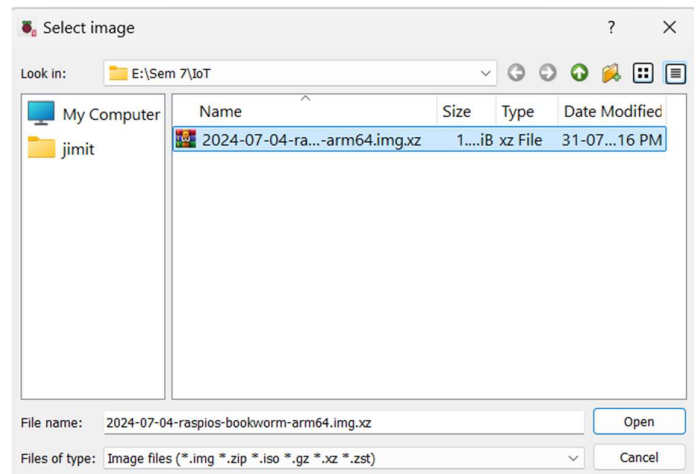
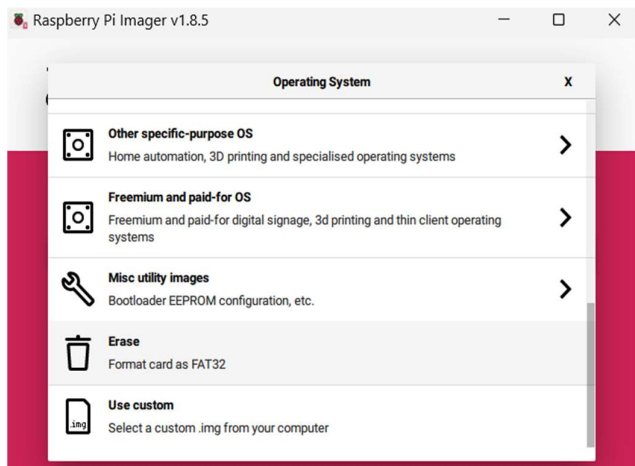
SSH (Secure Shell) is a protocol for secure remote login and command execution, essential for headless setups of the Raspberry Pi. Using the PuTTY tool, you can connect to the Raspberry Pi over the same network, using the hostname raspberrypi.local and port 22. VNC (Virtual Network Computing) allows remote control of the Raspberry Pi's desktop environment from a laptop, facilitated by the VNC Viewer tool. Connecting to raspberrypi.local, accepting the identity check and logging in with the Raspberry Pi's credentials provides visual interaction with its GUI.

Procedure:

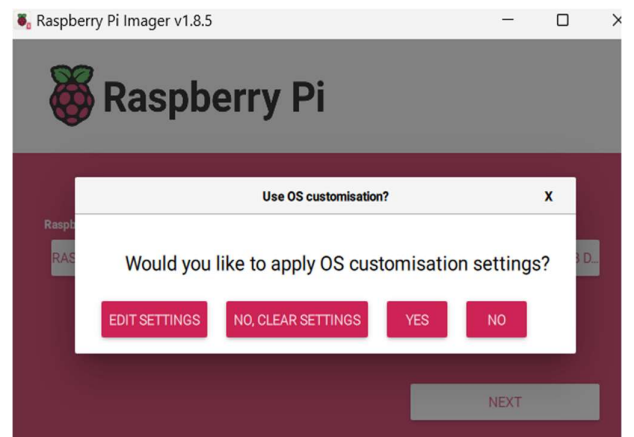
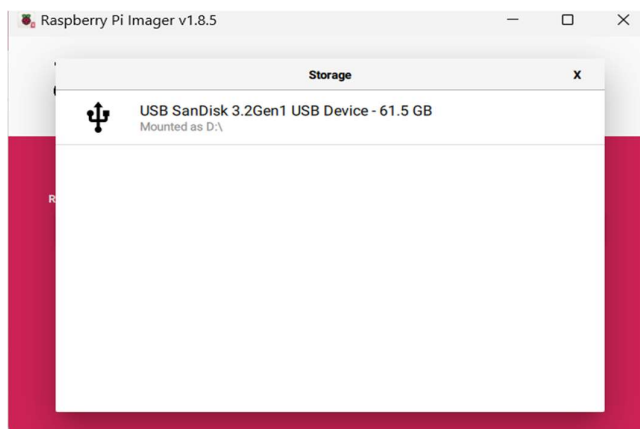
- 1) We will use the Raspberry Pi Imager for the flashing the OS in the Hard Drive. Download and install the Raspberry Pi Images, Putty Software, VNC Viewer and the Custom OS for the R Pi. Now, Open the Raspberry Pi Imager.

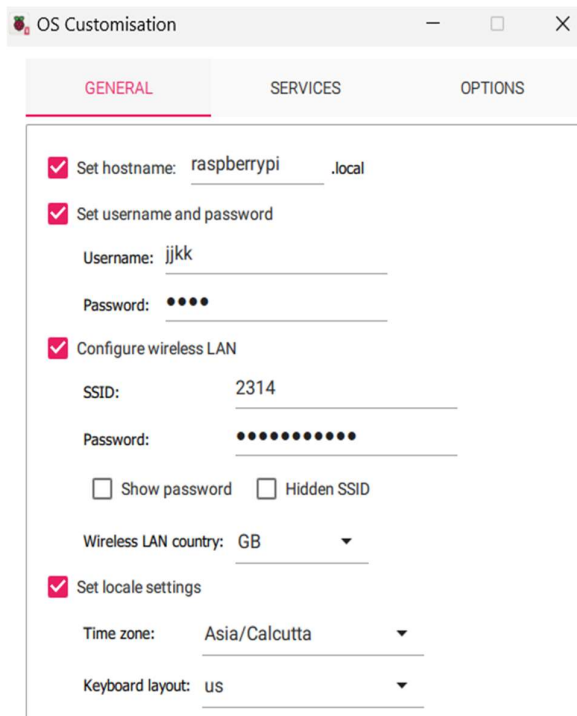


2) Click on the Choose Device and select the Raspberry Pi 4.



3) Choose the Custom OS and select the Downloaded Custom OS for the Raspberry Pi 4B. Plug in the Micro SD card or USB Drive (in our case). Choose the external storage option when prompted. Apply the OS customization and select the 'Edit Settings'.





OS Customisation

GENERAL SERVICES OPTIONS

☒ Set hostname: .local

☒ Set username and password

Username:

Password:

☒ Configure wireless LAN

SSID:

Password:

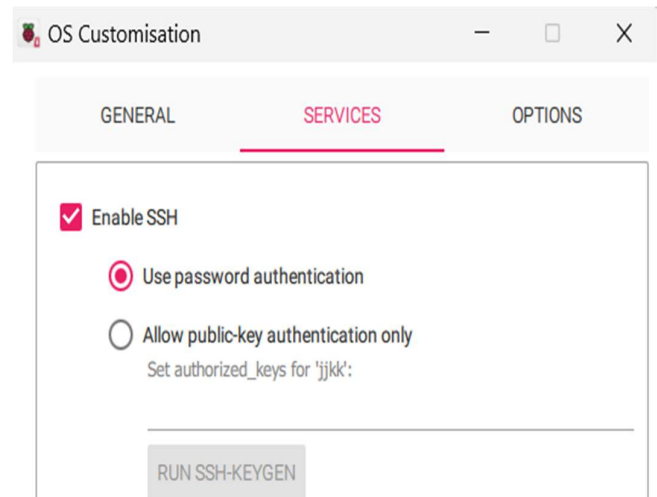
☐ Show password ☐ Hidden SSID

Wireless LAN country:

☒ Set locale settings

Time zone:

Keyboard layout:



OS Customisation

GENERAL SERVICES OPTIONS

☒ Enable SSH

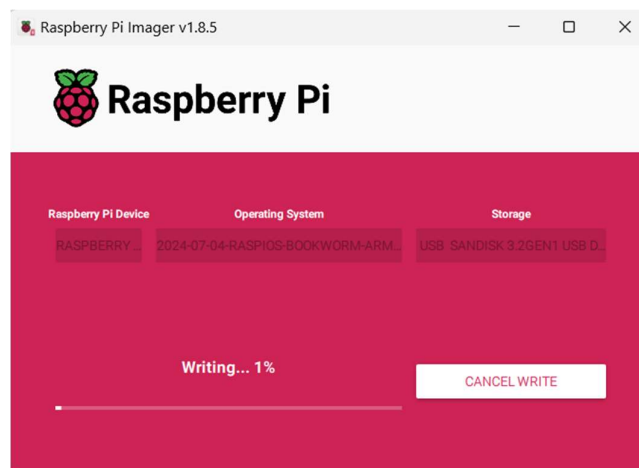
☒ Use password authentication

☐ Allow public-key authentication only

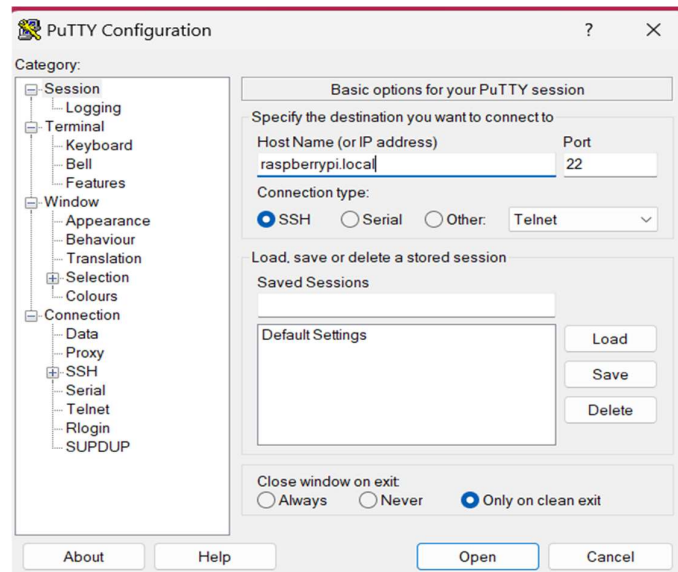
Set authorized_keys for 'jjkk':

RUN SSH-KEYGEN

- 4) Set the hostname which was by default as raspberrypi.local which will be used afterwards. Set the username and the password. Remember that we cant make any changes in the password in it. If forgotten we have to flash the entire OS again. Configure the WLAN preferably use you own mobile hotspot. Check the local settings. On Services open the Enable SSH which will be used afterwards. Save the changes and you are good to go. Flash it into the hard drive.



- 5) After plug the hard drive into the R Pi and give power supply gently wait for the 3-4 mins to get it booted and connect to the WLAN. Make sure the Led of the R Pi is steady. Now, Open the PuTTY Configuration panel and enter the hostname as raspberrypi.local by default port is 22 and hit the enter. Ensure the RPi and the Laptop is connected on the same network



- 6) After successful connection, Enter the password you created earlier. You should be able to log in to the Raspberry Pi successfully.

```

jjkk@raspberrypi: ~
login as: jjkk
jjkk@raspberrypi.local's password:
Linux raspberrypi 6.6.31+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.6.31-1+rpt1 (2024-05-29) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jul  4 05:47:41 2024
jjkk@raspberrypi:~ $

```

- 7) In the terminal console window type the `sudo raspi-config`. Then it will launch the configuration tool. Navigate to Interface Options using your keyboard's arrow keys and hit Enter. Go to VNC and hit Enter. Select Yes to confirm that you want to enable the VNC Server.

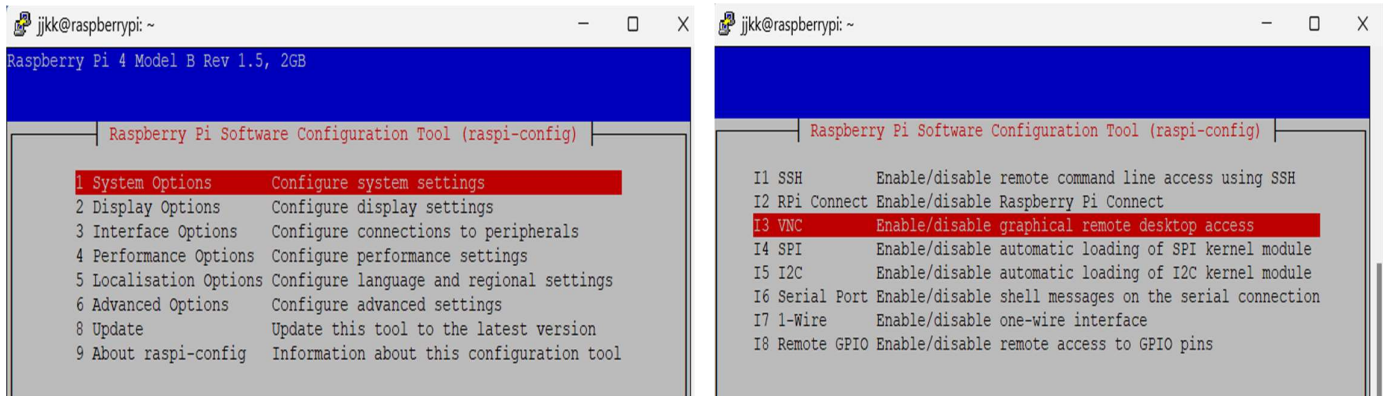
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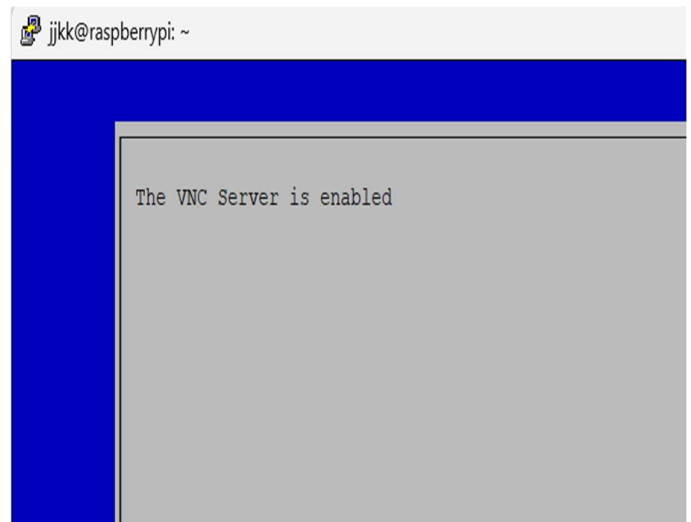
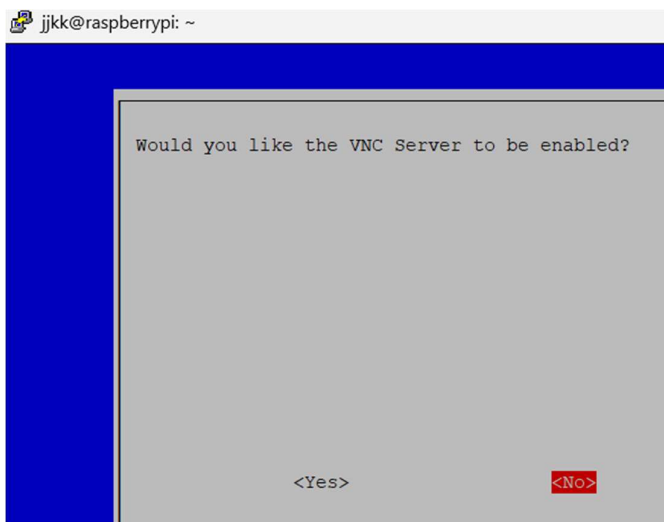
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permitted by applicable law.
Last login: Wed Aug  7 17:44:16 2024 from 2409:4080:961b:bad0:8cd9:c559:1f5a:cf3
jjkk@raspberrypi:~ $ sudo raspi-config

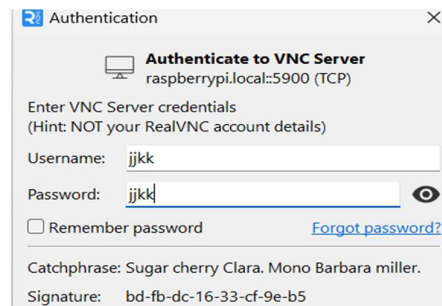
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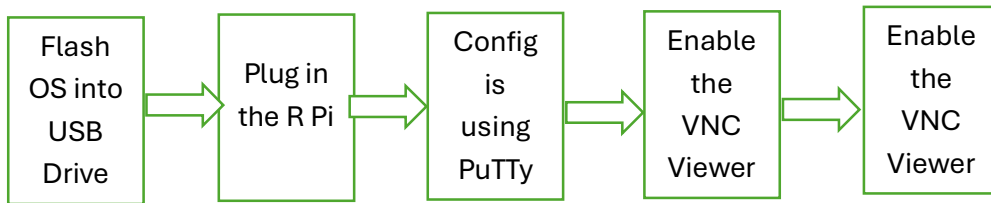


- 8) You can use your arrow key to navigate in the panel, Select third option which is Interface Options and hit enter, Now select the I3 which is Enable the VNC remote desktop access. Next will be Would you like the VNC Server to be enabled, Select Yes. When the VNC will be enabled successfully you be prompted as VNC Server is enabled.



- 9) Launch VNC Viewer, choose "Use VNC Viewer without signing in", enter "raspberrypi.local" in the search bar, click "Continue". If prompted enter your login credentials. In the Identity Check pop-up, click on Continue to connect to the Raspberry Pi board.

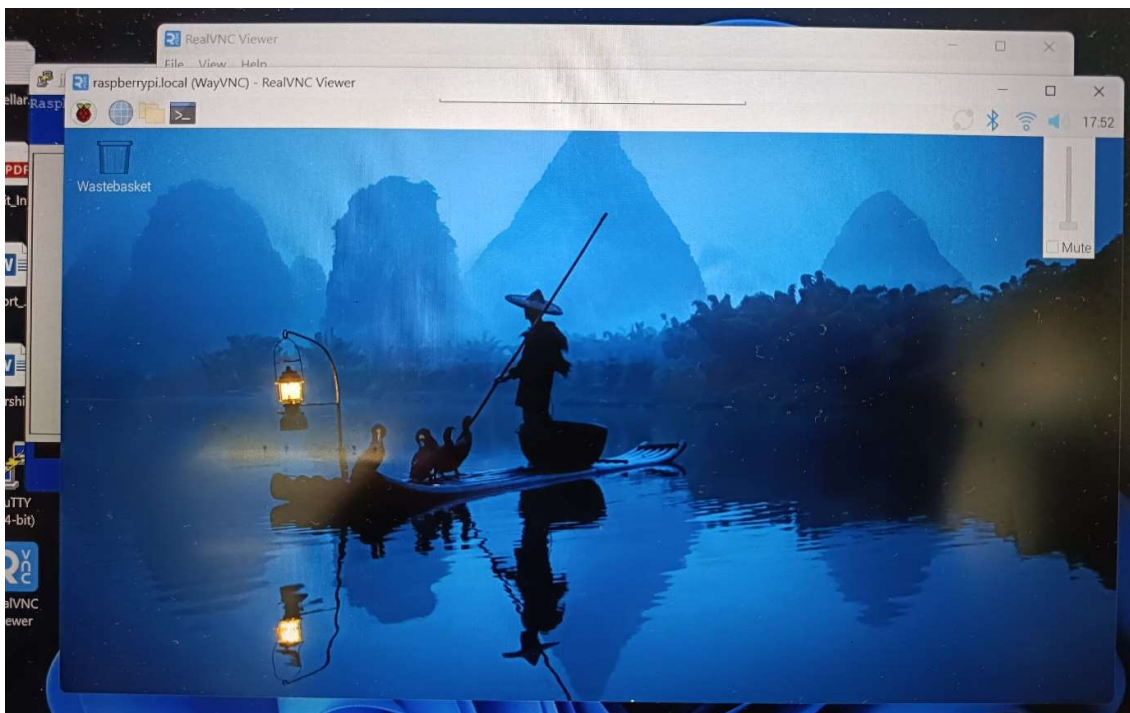


Block Diagram:

This block diagram provides a visual representation of the key steps involved in setting up a Raspberry Pi 4B with a custom OS, enabling SSH for secure remote login, and using VNC Viewer for remote desktop control.

Results:

The OS GUI of the Raspberry Pi 4 is successfully visible on the remote desktop through VNC Viewer.

**Discussion:**

This experiment demonstrates the process of setting up a Raspberry Pi 4B for remote access and control using SSH and VNC. By flashing a custom OS, configuring network settings, and enabling remote access protocols, we can manage and interact with the Raspberry Pi from another computer on the same network. This setup is particularly useful for headless operation scenarios where connecting a monitor, keyboard, and mouse to the Raspberry Pi is not feasible.

Conclusion:

The experiment successfully initialized and configured the Raspberry Pi 4B with a custom OS. The setup allowed secure remote access via SSH using PuTTY and remote desktop control using VNC Viewer. This configuration is practical for managing the Raspberry Pi in headless environments, enhancing its usability and flexibility in various applications.